

Tyler Johnston

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PROFESSIONAL SUMMARY

Full-stack developer based in Lisbon with 2 years of experience building production workflow applications using C#.NET, JavaScript/TypeScript, REST APIs, and Oracle SQL. MSc Information Management candidate at Nova IMS specializing in Business Intelligence, with experience across full-stack development, QA, production support, data-driven applications, and technical project delivery.

SKILLS

Programming C#, JavaScript, TypeScript, Python, Java, C++, Oracle SQL, Visual Basic, 4D
Frameworks & Tools .NET, REST APIs, React.js, Angular 17+, React Flow, Docker, Firebase, AWS, Power BI, Microsoft Fabric, Supabase, PostgreSQL
Data & ML Pandas, Plotly, YOLOv8, Roboflow, Monte Carlo simulation, geospatial datasets, Random Forest, clustering
DevOps & QA Git, Agile/Scrum, CI/CD, debugging, application testing, QA, production support
Languages English (Native), Portuguese (B1)

EXPERIENCE

- Full Stack Developer, Footprints, Inc. | Logan, UT** Mar 2024 - Aug 2025
- Built and maintained production workflow applications for aerospace and steel-manufacturing engineering teams using C#.NET, REST APIs, JavaScript, Ext JS, jQuery, Oracle SQL, and 4D.
 - Completed 240+ full-stack development tasks across Agile sprints, translating stakeholder requirements into front-end, back-end, database, API, and workflow automation features.
 - Served as a dedicated QA owner for full-stack releases, testing front-end, back-end, API, and database changes before deployment and resolving defects through root-cause analysis.
 - Optimized SQL queries, service-layer logic, and data retrieval workflows to improve reliability, maintainability, and performance across production systems.
- Computer Vision Research Assistant, Utah State University | Logan, UT** Jan 2024 - May 2024
- Built computer vision pipelines using YOLOv8 and Roboflow to detect curb ramps, pedestrian signage, and traffic lights in Google Street View imagery for ADA compliance analysis.
 - Automated image retrieval, GPS metadata extraction, and data preprocessing workflows to create geospatial datasets and heatmaps for city-scale accessibility evaluation.
- Modern C++ Teaching Assistant, Utah State University | Logan, UT** Aug 2023 - Dec 2023
- Designed a custom C++ std::vector prototype using templates, iterators, and manual memory management to optimize large-dataset allocation and demonstrate advanced data structure design.
- Web Developer, Space Dynamics Laboratory | Remote** Mar 2023 - Jun 2023
- Developed a full-stack workflow design tool using React, React Flow, C#.NET, and Docker microservices, enabling users to create process workflows through a responsive drag-and-drop interface instead of manual XML editing.
 - Enabled JSON and XML workflow configuration import/export through Docker-integrated services, reducing manual configuration errors and improving developer productivity.

PROJECTS

- Parallax Poker - Character-Driven Poker Roguelite** C# / Godot 4 / Python
- Built a C# / Godot 4 Texas Hold'em roguelite with a rules-correct poker engine covering all-ins, raise/reopening logic, uncalled chip refunds, O(1) hand evaluation, and asynchronous game-state management.
 - Developed a personality-driven AI decision engine with 15+ behavior parameters, equity calculations, draw detection, board texture scoring, SPR logic, tilt states, and probabilistic betting decisions.
 - Created a Monte Carlo balancing pipeline using headless Godot simulations, Python, Pandas, and Plotly to test 9 AI opponents across 288,000 simulated hands and identify win-rate outliers.
- Flashcard Dungeon - Full-Stack Language Learning App** Angular / TypeScript / Supabase
- Built a full-stack Angular 17+ and TypeScript language-learning app for European Portuguese, combining spaced-repetition flashcards, RPG dungeon combat, achievements, shop upgrades, and persistent progress tracking.
 - Designed a CEFR-mapped A1-B2 curriculum with directed node progression, prebuilt decks, pronunciation guidance from native Portuguese speakers, IndexedDB offline storage, and optional Supabase/PostgreSQL cloud sync.
- Absentee Analysis - HR Absenteeism Dashboard** Python / Vercel
- Segmented employees into absenteeism risk groups using K-Means, Hierarchical Clustering, and Gaussian Mixture Models; trained a Random Forest classifier to predict risk cluster membership.
 - Deployed an interactive HR analytics dashboard to Vercel, enabling workforce risk analysis through clustering, classification, and data visualization workflows.

EDUCATION

- Nova University, Lisbon, Portugal** May 2027
Master's in Information Management | Business Intelligence Specialization - Currently Enrolled
- Utah State University, Logan, Utah (USA)** May 2024
B.S. in Computer Science, Minor in Mathematics - Magna Cum Laude | GPA: 3.92 / 4.00 | Presidential Scholarship